

Machine Learning-Powered Network Anomaly Detection

An enterprise-grade, memory-safe framework designed to ingest high-velocity network flow telemetry, detecting DDoS floods and automated scans in real-time.

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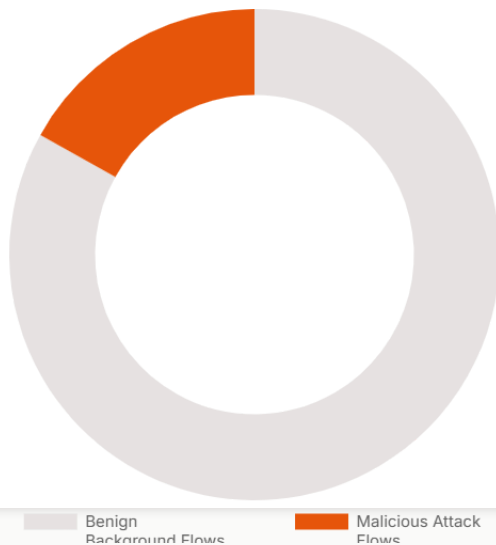
 Capstone Project Report

1. The Telemetry Landscape

The core operational challenge is the severe imbalance of network data. Malicious activity is a needle in a haystack. CyberFlow processes the UNB CIC-IDS2017 dataset using a 5% Stratified Downsampling Strategy to protect minority attack signatures.

Class Distribution (126,038 Flows)

Standard classifiers default to the majority class to minimize raw error rates. We optimize for F1-Score to counteract this.



The Packet Asymmetry Metric

Volumetric floods exhibit highly asymmetrical profiles (spraying packets with no replies). We engineered a dimensionless metric to isolate this footprint.

$$\text{Asymmetry} = (\text{Fwd_Pkts} - \text{Bwd_Pkts}) / (\text{Fwd_Pkts} + \text{Bwd_Pkts} + 10^{-5})$$

The Leakage Shield

Target proxy leakage causes artificially perfect training scores that fail in production. Our automated engine sweeps all 51 features to prevent "cheating".

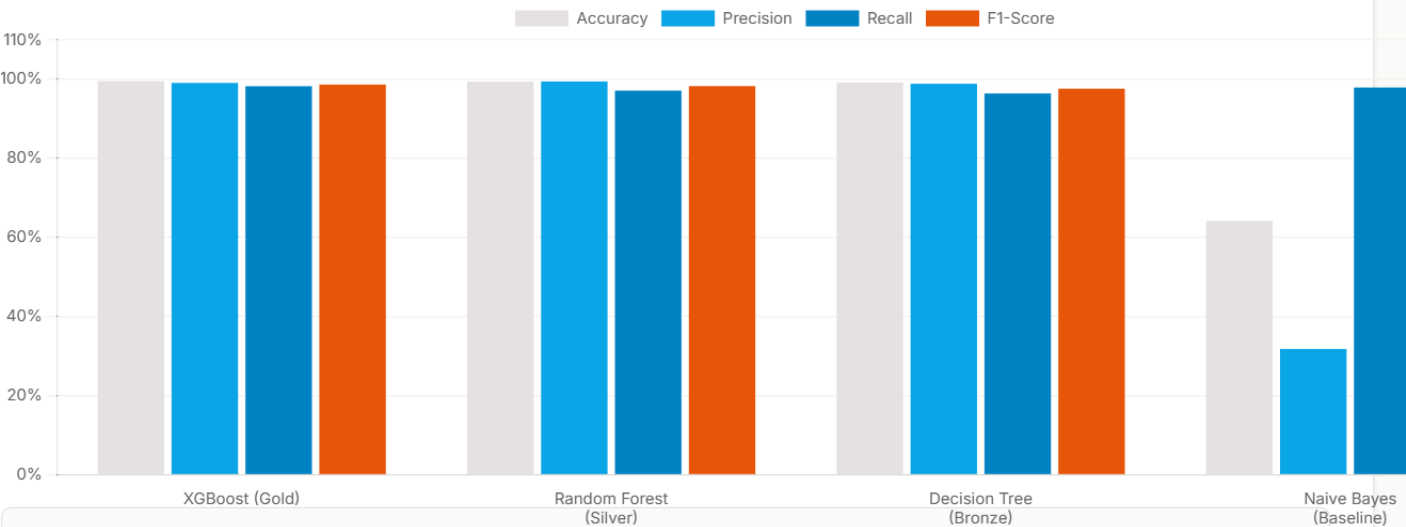
2. The Machine Learning Tournament

A four-way tournament evaluated distinct mathematical classifiers. To protect the SOC from alert fatigue, **Precision** and **F1-Score** were the primary KPIs.

Classifier Empirical Results

Evaluated on a 20% validation split (25,208 records).

Champion: XGBoost
Latency: 2.69s (Train) | <25ms (Inference)



Why XGBoost Won

Secured a peak F1-Score of 98.68% and Precision of 99.10%. Its sequential gradient boosting architecture minimizes error margins by continuously targeting misclassified residual weights.

Why Naive Bayes Failed

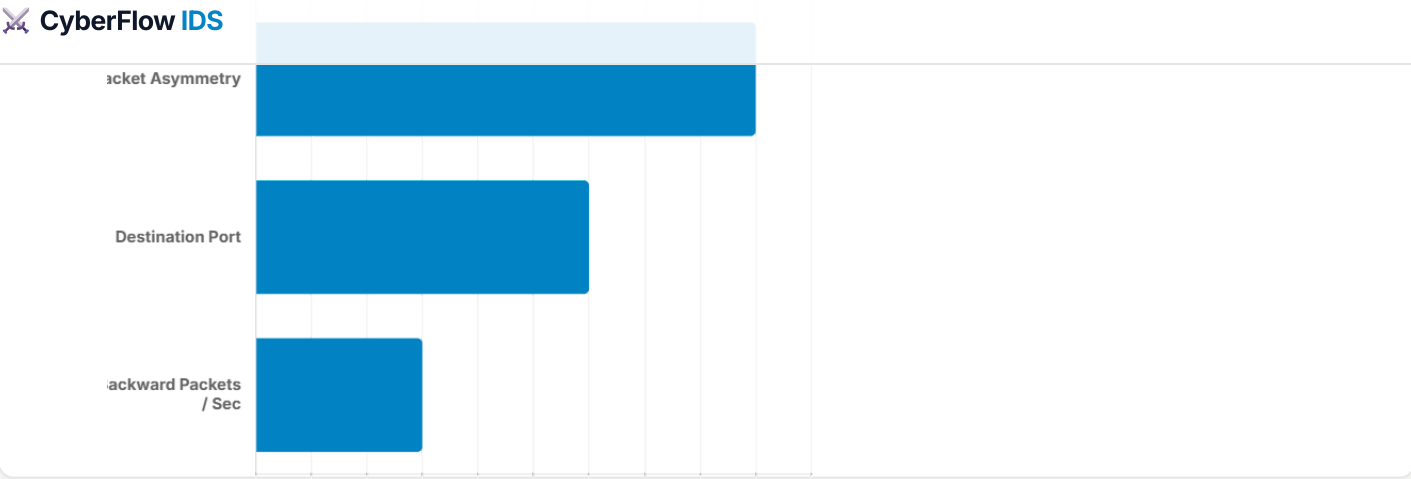
Yielded an unusable 31.81% precision. In network traffic, features (packet size, duration) are highly correlated. Naive Bayes' assumption of absolute independence was severely violated.

3. Technical Trust & Explainability

To avoid the risks of a "black-box" model, CyberFlow integrates cooperative game theory (SHAP) for explainability and enforces strict statistical parity audits to ensure safe automated blocking.

SHAP Feature Drivers

Marginal contribution of features across all possible combinations.



Algorithmic Fairness Audit

Auditing False Positive Rates (FPR) to prevent blocking critical background system protocols (e.g., DNS on 53, HTTPS on 443).

0.00%

System Ports
Ports ≤ 1023

0.00%

Dynamic Ports
Ports > 1023

Result: Absolute Statistical Parity Achieved.

4. Autonomous Operations & GenAI

Moving from an offline script to a FastAPI production service capable of gigabit line speeds (<25ms inference). To eliminate SOC alert fatigue, a GenAI Copilot (Gemini API) translates mathematical anomalies into human-readable mitigation briefs.

Live Simulation: GenAI SOC Copilot

Inject Anomalous Payload

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Waiting for telemetry input on port 8501... 
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